NEST: Locally Certified Search for Structural-Entropy Hierarchies

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Abstract. Structural entropy evaluates a graph hierarchy through the description length of an encoding tree. Existing methods such as HCSE and BBM construct such trees, but do not determine whether a returned hierarchy still admits an entropy-reducing local topology change. We introduce NNI-certified Entropy Search over Trees (Nest), a multistart optimizer based on rooted nearest-neighbor interchange (NNI). For weighted graphs, we derive the exact entropy change of one NNI and prove that an improving rotation can be inaccessible to a monotone single merge or compression. The identity supports best-improvement descent, a checkable one-NNI local-optimality certificate, and a bounded two-move search that crosses a controlled intermediate barrier while committing only a decreasing endpoint. We further prove that every multiway encoding tree has a binary refinement of no greater entropy. This yields an $O(3^n)$ structural-entropy specialization of subset-trellis inference and a global-optimum audit on small graphs. Across 50 graphs from five 64-vertex hierarchical regimes, Nest lowers entropy by 0.4290 ± 0.0281 bits relative to the better of HCSE and BBM and wins all paired comparisons. After development and calibration, a frozen 32-restart variant reaches the exact global optimum on 250/250 independently generated 12-vertex confirmation instances, versus 214/250 for the standard pool. These exact-instance results certify observed optimality without asserting a general approximation ratio.

Keywords: Structural entropy · hierarchical clustering · tree editing · nearest-neighbor interchange · graph algorithms

1 Introduction

Many networks are organized at more than one scale: vertices form local groups, related groups form larger modules, and the desired output is therefore a tree rather than one flat partition. Structural entropy supplies an information-theoretic objective for such an encoding tree [11]. Recent HCSE and BBM methods construct low-entropy hierarchies through recursive partitioning and balanced merging [16]. Their construction rules, however, do not answer a complementary question: *after a tree has been constructed, does a small topology change still lower the same objective?*

Construction and local optimality are distinct: early decisions can retain avoidable entropy. We therefore seek a topology operator whose effect is exact, whose termination is interpretable, and whose output can be compared with a true optimum on instances small enough for exhaustive certification.

Nearest-neighbor interchange (NNI) is the elementary rotation of a rooted binary tree. It changes one internal edge while preserving every leaf. NNI is standard in phylogenetic tree search and has been studied for other hierarchical-clustering objectives, but its structural-entropy change has not been used as an exact graph-hierarchy optimizer. We show that, for $((A, B), C) \to (A, (B, C))$, all objective terms cancel except two cross-module edge weights and three module volumes. This identity yields monotone descent and a one-NNI local-optimality certificate.

Our method, NNI-certified Entropy Search over Trees (Nest), refines several inexpensive initial trees with the exact operator and returns the lowest independently verified endpoint. A bounded two-move beam can cross a small intermediate barrier, but accepts a pair only when its endpoint lowers the full objective. A randomized-restart variant, Nest-R, increases basin coverage without inspecting labels or the exact optimum. We additionally specialize a subset-trellis dynamic program to compute the global optimum for small graphs. It separates three claims that are often conflated: monotone improvement, local certification, and global optimality.

Our contributions are:

- an exact weighted-graph NNI delta, a proof of separation from monotone single merge/compress steps, and a checkable one-NNI local certificate;
- Nest, which combines multi-start construction, exact NNI refinement, and bounded two-move escape without weakening the output guarantee;
- a binary-refinement theorem, an SE-specific $O(3^n)$ subset-trellis optimizer for small graphs, and an exact audit of Nest's optimality gap; and
- a paired evaluation against SE agglomeration, Louvain-2L, Paris, and the official HCSE and BBM implementations, with entropy, hierarchy recovery, runtime, memory, operator audits, and component ablations.

Across five frozen regimes, Nest lowers entropy by 0.5512 ± 0.0632 bits versus HCSE and 0.5175 ± 0.0319 bits versus BBM, winning every paired comparison. On a frozen 250-graph confirmation suite, Nest-R with 32 random starts reaches the global optimum in 250/250 cases, compared with 214/250 for the standard candidate pool. This is an empirical finite-suite certificate, not a worst-case approximation guarantee.

2 Preliminaries

Weighted graph and encoding tree. Let $G = (V, E, w)$ be an undirected graph with nonnegative edge weights. For $S \subseteq V$, let $V_S = \mathrm{vol}(S) = \sum_{u \in S} d_u$, and let g_S be the total weight of edges with exactly one endpoint in S. The total graph volume is $\mathrm{vol}(V) = \sum_u d_u = 2 \sum_{e \in E} w_e$. An encoding tree T has one leaf per vertex. Each internal node a represents the union S_a of its descendant leaves; a^- denotes its parent.

The structural entropy of G under T is

$$H^T(G) = -\sum_{a \neq r} \frac{g_{S_a}}{\mathrm{vol}(V)} \log_2 \frac{V_{S_a}}{V_{S_{a^-}}}. \tag{1}$$

Our sole optimization objective is Eq. (1); lower is better. No ground-truth label is used to construct or select a tree. For a zero-volume module, its summand is defined as zero; such a module has no incident positive-weight edge and can be attached after optimizing the positive-volume part of the tree.

Constructors. We distinguish a *constructor*, which produces an initial tree, from a *refiner*, which searches neighboring topologies. Our candidate pool contains (i) a binary SE-agglomerative dendrogram, (ii) a recursive SE partition tree, and (iii) a Paris dendrogram when the optional implementation is available [2]. HCSE and BBM are distinct external constructors and are evaluated from their official implementation [16].

Rooted NNI. An NNI move acts on a binary parent–child edge. Writing the local topology as $((A, B), C)$, the two nontrivial alternatives are $(A, (B, C))$ and $(B, (A, C))$. The leaf set, graph, and all modules outside the local parent $P = A \cup B \cup C$ remain unchanged. A rooted binary tree with n labeled leaves has finitely many topologies, namely $(2n-3)!!$, so any strictly decreasing local search terminates.

3 NNI-Certified Entropy Search

Nest constructs several candidate hierarchies and refines each in a canonical binary neighborhood. Its local effect is exact, every committed endpoint is independently rescored, and small instances admit a separate global certificate.

3.1 A local structural-entropy identity

For disjoint sets X, Y, write $W(X, Y) = \sum_{u \in X, v \in Y} w_{uv}$. Consider the rotation shown in Fig. 1. Let $X = A \cup B$, $Y = B \cup C$, and $P = A \cup B \cup C$.

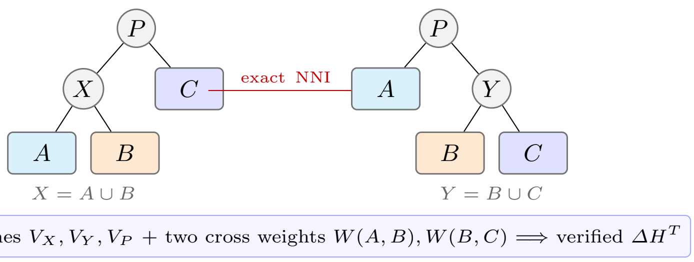

Fig. 1. A rooted NNI replaces only $X = A \cup B$ by $Y = B \cup C$. The highlighted local statistics determine the exact full-objective change; all leaves and the surrounding tree remain fixed.

Theorem 1 (Exact weighted NNI delta). *For an eligible rotation with $V_X, V_Y, V_P > 0$, the change in tree structural entropy is*

$$\Delta H^T = \frac{2}{\mathrm{vol}(V)} \left[W(A,B) \log_2 \frac{V_P}{V_X} + W(B,C) \log_2 \frac{V_Y}{V_P} \right]. \tag{2}$$

The symmetric alternative follows by exchanging A and B.

Proof. Deferred to the optional supplementary appendix.

Proposition 1 (Atomic advantage over merge–compress). *Let T_c contract the edge from P to $X = A \cup B$, exposing A, B, C as siblings, and let T' then merge B, C under $Y = B \cup C$. For these HCSE compress and merge primitives,*

$$H^{T_c}(G) - H^T(G) = \frac{2W(A,B)}{\mathrm{vol}(V)} \log_2 \frac{V_P}{V_X} \geq 0, \tag{3}$$

$$H^{T'}(G) - H^{T_c}(G) = -\frac{2W(B,C)}{\mathrm{vol}(V)} \log_2 \frac{V_P}{V_Y} \leq 0. \tag{4}$$

Their sum is Eq. (2). Thus NNI lowers entropy when the merge gain in Eq. (4) exceeds the compression loss in Eq. (3), although neither desired reparenting nor an improving compression is available as one primitive.

Proof. Deferred to the optional supplementary appendix.

This proves strict separation from monotone single-primitive search, not global dominance over a complete HCSE round, which may cross an intermediate increase. We evaluate Eq. (2) from cached sets and sparse adjacency lists, then verify every selected move by fully recomputing Eq. (1).

3.2 One-step certificate and compound escape

Algorithm 1: Exact NNI refinement of one candidate tree

Input: Graph G, tree T, tolerance ε, barrier β, beam b

repeat
  evaluate Eq. (2) for every legal NNI of T;
  apply the most negative move and verify the full objective;
until *no improving one-step move exists*;
for *at most R compound rounds* **do**
  retain the b first moves with smallest ΔH and $\Delta H \leq \beta$;
  expand every legal second NNI from each retained intermediate tree;
  if *no two-move endpoint has entropy below $H^T(G)$* **then**
    break;
  commit the best endpoint; rerun one-step descent;
return T

Proposition 2 (Monotonicity and local certificate). *One-step refinement never increases $H^T(G)$. If it stops before its move budget, the returned tree has no improving rooted NNI on any eligible binary parent–child edge.*

Proof. Deferred to the optional supplementary appendix.

Strict one-step descent can stop before a better tree separated by a shallow barrier. Algorithm 1 therefore keeps a bounded beam of first moves whose increase is at most β and evaluates a second NNI. It never commits the intermediate tree alone.

Proposition 3 (Safe compound search). *The compound stage never worsens its input. It can reach an improving endpoint even when every one-step neighbor is non-improving.*

Proof. Deferred to the optional supplementary appendix.

3.3 Fast NNI-certified entropy search

Nest applies Algorithm 1 independently to SE-agglomerative, recursive-SE, and Paris starts and returns the candidate with minimum verified entropy. The initializer is recorded for every output, so gains can be attributed to candidate selection, one-step refinement, and compound escape. With n leaves there are at most $2(n-2)$ oriented NNI alternatives per sweep. Our implementation uses best improvement, cached leaf sets, sparse cross-weight scans, and an $O(m \log h + n)$ full verification only for a selected move, where h is tree height. We make runtime claims empirically rather than assuming balanced trees.

Randomized basin coverage. Nest-R adds q coalescent starts, each built by repeatedly merging two uniformly selected current components, then applies the same NNI routine. The distribution is not uniform over labeled topologies. Selection uses only rescored entropy; increasing q cannot worsen the result, but gives no global guarantee.

3.4 Exact global audit on small graphs

Local certification does not imply global optimality. We therefore use a separate exact solver. First, binary trees suffice.

Proposition 4 (Binary sufficiency). *Every encoding tree has a binary refinement whose structural entropy is no greater. Consequently, a global minimum exists among rooted binary trees.*

Proof. Deferred to the optional supplementary appendix.

For a nonempty vertex subset S, let $F(S)$ be the minimum contribution of a binary subtree rooted at S, excluding the edge from S to its parent. Then

$$\begin{aligned} &F(\{v\}) = 0, \\ &F(S) = \min_{A \dot\cup B = S} \left\{ F(A) + F(B) - \frac{g_A}{\mathrm{vol}(V)} \log_2 \frac{V_A}{V_S} - \frac{g_B}{\mathrm{vol}(V)} \log_2 \frac{V_B}{V_S} \right\}. \end{aligned} \tag{5}$$

Enumerating each unordered split once gives $O(3^n)$ time and $O(2^n)$ memory, an SE-specific instance of subset-trellis exact inference [13]. By Proposition 4, $F(V)$ is the global minimum over all encoding trees, not only over the binary subclass. This exponential routine is used only to audit 12-vertex instances; it is not part of Nest at deployment scale.

For an additional graph-only certificate, an edge-LCA expansion gives

$$H^T(G) = \sum_{uv \in E} \frac{w_{uv}}{\mathrm{vol}(V)} \log_2 \frac{V^2_{\mathrm{lca}(u,v)}}{d_u d_v} \geq \sum_{uv \in E} \frac{w_{uv}}{\mathrm{vol}(V)} \log_2 \frac{(d_u + d_v)^2}{d_u d_v}. \tag{6}$$

The inequality follows from $V_{\mathrm{lca}(u,v)} \geq d_u + d_v$. It is a universal lower bound, but the exact dynamic program gives the sharper certificates reported below.

4 Experiments

4.1 Questions and protocol

We ask: (Q1) how often do established constructors leave an improving NNI; (Q2) does compound search add improvements beyond a one-step certificate; (Q3) does lower entropy preserve or improve recovery of a planted hierarchy; and (Q4) how close is Nest to the global optimum on exactly solvable graphs?

The frozen main protocol uses ten graph seeds per regime. Every method sees the identical weighted graph, and every tree is rescored by the same independent implementation of Eq. (1). We report paired means and 95% t-intervals, raw per-seed records, wall-clock time, and Python allocation peak. The compound search uses beam width 16, barrier $\beta = 0.05$ bits, and at most eight accepted compound rounds. No label is used by Nest for model selection. BBM is additionally shown with the planted number of fine clusters, which is an oracle advantage and is marked as such.

Data. Our controlled suite contains five two-level hierarchical stochastic block models: clean, noisy, imbalanced, weighted, and weak-hierarchy. Each balanced graph has 64 vertices, eight fine blocks, and four coarse blocks; the imbalanced regime has the same total size with block sizes from 4 to 12. The generator manifest, any 0.01-weight connectivity edges, and every seed are stored with the results. This family follows the hierarchical stochastic block model used to study recovery beyond worst-case inputs [5].

Methods. We compare SE agglomeration, a two-level Louvain tree, Paris [2], official HCSE, official BBM with oracle k [16], and Nest. We also refine each external constructor independently to measure its one-step and compound gap rather than crediting all gains to our preferred initializer.

Table 1. Tree structural entropy H^T (mean ± 95% CI; lower is better). BBM† receives the planted fine-cluster count.

Method	Clean	Noisy	Imbal.	Weighted	Weak
SE–agglomerative	2.920 ± 0.094	3.712 ± 0.070	3.218 ± 0.116	2.444 ± 0.079	3.424 ± 0.112
Louvain–2L	4.167 ± 0.066	4.575 ± 0.061	4.270 ± 0.060	3.647 ± 0.145	4.425 ± 0.095
Paris	2.919 ± 0.112	3.698 ± 0.072	3.165 ± 0.106	2.348 ± 0.072	3.433 ± 0.112
HCSE	3.300 ± 0.190	4.351 ± 0.170	3.685 ± 0.122	2.759 ± 0.130	4.040 ± 0.286
BBM†	3.359 ± 0.122	4.062 ± 0.073	3.634 ± 0.124	2.996 ± 0.073	3.916 ± 0.064
NEST (ours)	**2.845 ± 0.094**	**3.675 ± 0.067**	**3.135 ± 0.115**	**2.337 ± 0.073**	**3.387 ± 0.104**

Metrics. The primary metric is $H^T(G)$. Dendrogram purity is computed for the planted fine and coarse partitions: for every same-class vertex pair, it measures the class fraction under the pair's predicted LCA, then averages over pairs. We also report the fraction of runs improved by one-step and compound NNI, number of moves, runtime, and memory. For the exact suite we report the additive gap $H^T_{\mathrm{NEST}} - H^*$, relative gap $(H^T_{\mathrm{NEST}}/H^* - 1)100\%$, and *optimal-hit rate*: the fraction of graph instances whose objective is within $\max(10^{-9}, 10^{-8}|H^*|)$ of the exact optimum. This objective metric does not assert recovery of a unique planted tree.

4.2 A strict one-step trap

Before aggregate evaluation, an eight-vertex weighted witness verifies the purpose of compound search. Exact one-step descent stops at 1.970038 bits. Every one-NNI delta is nonnegative, but a first move crossing a 0.013970-bit barrier followed by a second move and ordinary descent reaches 1.920593 bits, a 0.049445-bit reduction. This witness is generated from a fixed seed and is covered by a regression test.

4.3 Frozen benchmark results

Table 1 reports the raw output of each method, before applying our audit to competing trees. NEST has the lowest mean H^T in all five regimes and on 50/50individual paired graphs. Its paired reduction is 0.5512±0.0632 bits relative to HCSE and 0.5175±0.0319 bits relative to BBM; it is lower on all 50 graphs against each method. Figure 2 shows paired evidence against the stronger of HCSE and BBM on each graph.

4.4 The operator audit

The constructor audit in Table 2 asks whether a returned tree can be improved by the exact local operator. One-step NNI improves SE agglomeration and oracle BBM on every graph. After one-step descent has certified its local neighborhood, bounded compound search still improves 72% of SE-agglomerative trees and 92% of BBM trees. Paris is closer to local optimality but is not immune. For HCSE, one-step NNI lowers the imbalanced seed-5 tree by 0.001352 bits, while compound search lowers the weighted seed-2 tree by 0.000637 bits. These witnesses support

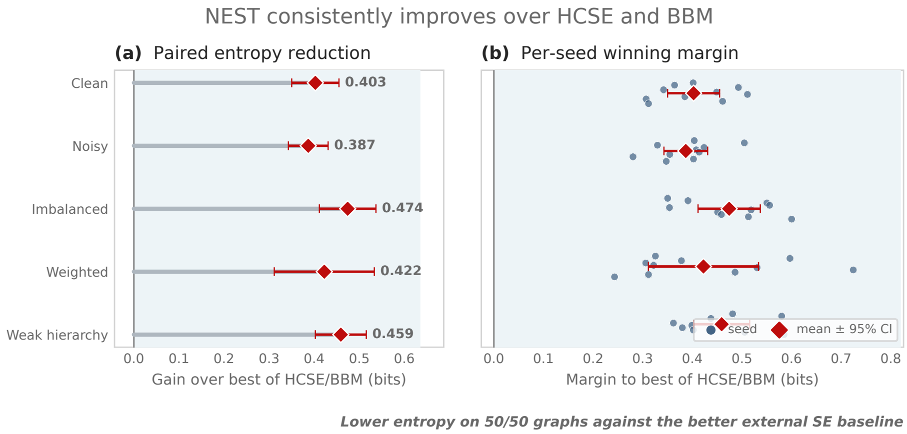

Fig. 2. Paired advantage over external structural-entropy methods. For each graph, the reference is the lower-entropy result of HCSE and BBM. (a) Regime-level mean reduction and 95% paired confidence interval. (b) Per-seed margins with small vertical jitter for visibility. NEST has lower entropy on all 50 graphs.

Table 2. NNI audit pooled over 50 paired graphs. Reduction is relative to each constructor's raw H^T; rates are the fractions of runs improved.

Constructor	1-NNI gain	1-NNI hit	2-step gain	2-step hit	Time
SE–agglomerative	1.99%	100%	0.15%	72%	0.334s
Paris	0.06%	68%	0.07%	38%	0.181s
HCSE	0.00%	2%	0.00%	2%	0.055s
BBM	13.98%	100%	0.47%	92%	1.038s

the neighborhood-separation claim without asserting that HCSE is generally weak.

4.5 Components and exact optimality

NEST forms a small candidate set from SE agglomeration, recursive SE, and Paris, refines each with the same exact operator, and returns the lowest independently rescored endpoint. The main benchmark contains 350 method records and 50 generator manifests. Its verifier checks schema, paired seeds, unique keys, and monotonicity of committed NNI endpoints.

Table 3 separates the three algorithmic choices. The candidate pool improves 34 of 50 graphs over SE agglomeration; exact one-step NNI then improves 45 of 50 paired multi-start outputs; bounded compound search adds a smaller but nonzero gain on 30 graphs. Hence neither the initializer pool nor the escape stage alone explains the full result.

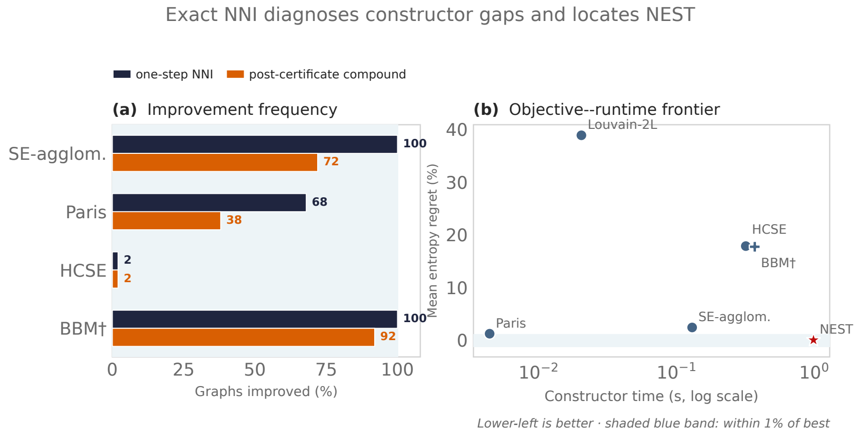

Fig. 3. Operator diagnostics pooled over 50 paired graphs. (a) Fraction of external constructor outputs improved by one-step NNI and by compound search after the one-step certificate. (b) Mean entropy regret versus constructor runtime; lower-left is better. Runtime includes Python allocation tracing and should be interpreted comparatively.

Table 3. Component ablation pooled over 50 paired graphs. The gain and improved-run columns compare each row to the preceding row.

Variant	Mean H^T	Mean gain	Improved runs
SE agglomeration	3.1435	–	–
+ candidate pool	3.1041	0.0394	34/50
+ exact 1-NNI	3.0794	0.0247	45/50
+ compound escape	3.0759	0.0035	30/50

For a global audit, we generate five analogous 12-vertex HSBM regimes. The initial 50 graphs (seeds 0–9) expose five misses by the standard candidate pool: mean relative gap 0.118% and worst gap 3.72%. Four, eight, and 16 starts raise hit rate from 90% to 96%, 98%, and 100%. A 250-graph calibration (seeds 10–59) leaves three misses at 16 starts, fixing $q = 32$. A second split (60–109) gives 250/250 hits. Since the original protocol reused seeds across regimes, the final confirmation (110–159) uses disjoint streams: NEST-R32 reaches 250/250, versus 214/250 for the standard pool (Table 4). Totals over 800 graphs are 800 and 685. This establishes neither a size-independent approximation ratio nor guaranteed global optimality beyond the audited instances.

4.6 Real networks

Table 5 complements the controlled suites with four networks bundled by NetworkX or the artifact library. NEST obtains the lowest raw entropy on 4/4networks. Because only Karate has a standard class label in the bundled loader and the

Table 4. Exact audit on 12-vertex HSBMs. R32 was frozen after calibration. The final confirmation uses disjoint RNG streams across regimes. Hit means equality to H^* within the stated tolerance.

Split	Graphs	Standard hit	NEST-R32 hit
Development (0–9)	50	45 (90.0%)	50 (100%)
Calibration (10–59)	250	211 (84.4%)	250 (100%)
Confirmation A (60–109)	250	215 (86.0%)	250 (100%)
Confirmation B (separate RNG)	250	214 (85.6%)	**250 (100%)**
Overall	800	685 (85.6%)	**800 (100%)**

Table 5. Tree entropy on four bundled real networks (lower is better). BBM uses the ground-truth k when labels exist and Louvain's k otherwise.

Method	Karate	Florent.	Les Mis.	Davis
SE–agglomerative	2.698	1.923	3.078	3.055
Louvain–2L	3.405	2.452	3.923	3.813
Paris	2.597	1.923	2.920	2.981
HCSE	2.929	2.172	3.930	3.206
BBM	3.330	2.254	3.199	3.387
NEST (ours)	**2.586**	**1.919**	**2.916**	**2.962**

remaining networks lack a unique ground-truth hierarchy, this experiment tests objective minimization rather than recovery.

5 Related Work

Structural entropy and hierarchy construction. Li and Pan introduced structural information as an encoding-tree measure of network organization [11]. HCSE and BBM develop hierarchy construction and balancing operations for this objective [16]. The structural-entropy survey organizes the broader theory, optimization, and application literature [18]; HypCSE subsequently develops a continuous hyperbolic formulation for hierarchical clustering [19]. NEST asks whether a returned tree is locally optimal under an exact topology rotation and uses the answer as both a diagnostic and an optimizer.

Graph hierarchical clustering. Classical hierarchical clustering is predominantly agglomerative or divisive [15]. Paris is an agglomerative graph hierarchy related to modularity and provides a strong inexpensive initialization [2]. Dasgupta-style objectives initiated a complementary algorithmic literature [7]; average linkage and divisive local search have guarantees for its revenue dual [14]. Sparsest-cut and spreading-metric algorithms give approximations for that family [3, 17], while admissible hierarchical objectives have been characterized more generally [6]. Continuous ultrametric fitting offers another optimization route [4]. These objectives are useful conceptual neighbors, but a local delta derived for one cannot be substituted for Eq. (1).

Exact inference and heuristic accuracy. Cluster trellises and trellis-guided A* support exact hierarchical inference [13, 9]. Our routine is an SE specialization: Proposition 4 justifies the subset recurrence. Exact-optimum hit rate has assessed k-means and median-tree heuristics [1, 8]; here it complements planted-hierarchy recovery.

Local search in tree space. NNI is a classical neighborhood for phylogenetic trees; its reconfiguration distance is itself nontrivial [12]. Recent work studies NNI local search for hierarchical-clustering costs [10]. Our formula targets weighted graph structural entropy and exposes the two cross-module weights controlling a rotation. Unlike unit-clique landscape analyses, our method evaluates the complete objective on arbitrary weighted graphs. Its new content is the verified optimizer, compound search, multi-start construction, comparative evaluation, and exact global audit; local certification alone is not global optimality.

6 Limitations and Conclusion

NNI is a local operator, not a certificate of global optimality. Compound search explores only a bounded two-step neighborhood and does not exhaust all encoding trees. Multiway regions must also be resolved by a constructor before binary rotations can act on them. Lower structural entropy need not improve every external notion of hierarchy, which is why we report planted purity alongside the optimized objective. The exact solver is exponential and its near-optimality measurements therefore cover only the declared 12-vertex suite.

We introduced Nest, which derives an exact weighted NNI identity and uses it for monotone refinement, a local-optimality certificate, and safe bounded escape from shallow one-step traps. On the frozen five-regime suite, Nest achieved the lowest mean entropy in every regime and outperformed both HCSE and BBM on all 50 paired graphs. A binary-refinement theorem and exact subset dynamic program additionally show that Nest reaches the global optimum on 250/250 confirmation instances, versus 214/250 for the standard pool, on the audited 12-vertex HSBM family. This does not imply global optimality beyond the audited suite. The operator audit further shows that hierarchy construction, local optimality, and global optimality are distinct concerns. The result is a reproducible hierarchy-search framework whose guarantees and empirical gains can be checked directly.

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A Optional Appendix: Proof Details

Proof of Theorem 1. First suppose the three child modules have positive volume; the zero-volume case follows by deleting modules with no incident positive-weight edge and taking the zero-contribution limit. Only the terms for A, B, C, X, and Y can change. Omitting the common factor $1/\operatorname{vol}(V)$, their old and new contributions are

$$\begin{aligned} L_{\text{old}} &= g_A \log_2 \frac{V_X}{V_A} + g_B \log_2 \frac{V_X}{V_B} + g_X \log_2 \frac{V_P}{V_X} + g_C \log_2 \frac{V_P}{V_C}, \\ L_{\text{new}} &= g_A \log_2 \frac{V_P}{V_A} + g_B \log_2 \frac{V_Y}{V_B} + g_C \log_2 \frac{V_Y}{V_C} + g_Y \log_2 \frac{V_P}{V_Y}. \end{aligned}$$

Using $g_X = g_A + g_B - 2W(A,B)$ and $g_Y = g_B + g_C - 2W(B,C)$, the coefficients of g_A, g_B, g_C telescope to zero. The remaining terms are

$$2W(A,B)\log_2(V_P/V_X) + 2W(B,C)\log_2(V_Y/V_P).$$

Restoring $1/\operatorname{vol}(V)$ proves Eq. (2). External boundary weights and $W(A,C)$ do not enter the delta.

Proof of Proposition 1. Delete X's codeword and substitute $g_X = g_A + g_B - 2W(A,B)$ in Eq. (1); this gives Eq. (3). The symmetric calculation for inserting Y gives Eq. (4).

Proof of Proposition 2. Only moves with $\Delta H^T < -\varepsilon$ are committed, and their endpoint is independently verified. At termination every enumerated legal move has $\Delta H^T \geq -\varepsilon$.

Proof of Proposition 3. A pair is committed only after its final tree is fully recomputed and found to have lower entropy than the tree preceding the pair. The first move is not committed independently, so its sign cannot weaken the output guarantee. A two-edge path may have a positive first delta and a negative net delta; the regression witness in Sec. 4 instantiates this case.

Proof of Proposition 4. At a node P, choose two children with disjoint leaf sets A, B, insert their union $S = A \cup B$ below P, and make A, B children of S. The objective change is

$$-\frac{2W(A,B)}{\operatorname{vol}(V)} \log_2 \frac{V_P}{V_S} \leq 0.$$

Repeated insertion converts every multiway node to binary without increasing the objective.

A.1 Exact and sampled basin probabilities

The finite hit rates in the main paper do not supply a per-start probability. For a fixed graph G, deterministic refinement map F, optimum H^*, and start measure μ, that probability is

$$p_G = \sum_T \mu(T)\, \mathbf{1}\{H^{F(T)}(G) = H^*(G)\}.$$

It depends on the graph, start distribution, tie-breaking, and refinement configuration. The exact subset DP supplies H^* but not p_G.

For $n \leq 8$ we enumerate all $(2n-3)!!$ unordered rooted binary labeled topologies. Uniform topology mass is not NEST's restart distribution. Under the pairwise coalescent, a topology with child subtrees of a, b leaves has compatible-history count

$$h(T) = \binom{a+b-2}{a-1} h(T_A)h(T_B), \qquad \mu(T) = \frac{h(T)}{\prod_{k=2}^{n} \binom{k}{2}}.$$

The history counts sum exactly to the denominator. Table 6 shows that the selected hard $n = 8$ graph has exact coalescent basin mass $p_G = 0.252365$; because the standard pool misses it, 32 independent starts give $1-(1-p_G)^{32} = 0.999909$, not certainty.

Table 6. Exact basin mass on one fixed noisy HSBM graph at each size. p_U uses the uniform measure over unordered rooted binary labeled topologies; p_C uses NEST's pairwise-coalescent measure. "Strict" requires every planted fine and coarse block to occur as a clade.

n	starts	pool hits?	$p_U(H^*)$	$p_C(H^*)$	p_C(strict)
5	105	yes	100.000%	100.000%	0.000%
6	945	yes	100.000%	100.000%	0.000%
7	10,395	yes	100.000%	100.000%	0.000%
8	135,135	no	24.848%	25.237%	0.000%

At $n = 12$, exact enumeration would require 13,749,310,575 topologies. Table 7 therefore uses direct Bernoulli sampling rather than the earlier first-hit diagnostic. The estimates vary substantially by graph: 8.07%–46.43% per start. On noisy-11 this implies only 93.23% for 32 random starts (a conservative 95% lower value of 91.87%), explaining why the observed finite-suite success must not be read as a probabilistic guarantee.

Table 7. Direct Monte Carlo basin estimates on the eight preidentified hard $n = 12$ graphs (10,000 independent coalescent starts each). Intervals are exact 95% Clopper–Pearson intervals. $q_{32} = 1 - (1 - \hat{p})^{32}$; its lower column uses the lower confidence limit for p. Strict recovery was observed 0/10,000 times on every graph (per-graph 95% upper bound 0.0369%).

Graph	$\hat{p}(H^*)$	95% CI	q_{32}	lower q_{32}
noisy-0	17.19%	[16.46, 17.94]%	99.761%	99.683%
noisy-8	14.61%	[13.92, 15.32]%	99.362%	99.175%
noisy-9	20.56%	[19.77, 21.37]%	99.937%	99.913%
imbal.-0	46.43%	[45.45, 47.41]%	100.000%	100.000%
weak-9	30.61%	[29.71, 31.52]%	99.999%	99.999%
noisy-11	8.07%	[7.54, 8.62]%	93.229%	91.872%
noisy-17	17.75%	[17.01, 18.51]%	99.808%	99.743%
weak-17	17.86%	[17.11, 18.63]%	99.816%	99.754%

Objective success is deliberately separated from strict planted recovery. No one of the 80,000 hard-case endpoints contained every planted fine and coarse block as a clade; the exact $n = 8$ enumeration likewise gives zero mass. This strict event is stronger than dendrogram purity and exposes objective– recovery misalignment on these small realized HSBMs. Thus the restart evidence supports optimization of H^T, not a claim that every entropy optimum equals the generator's planted hierarchy.